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1  import threading
2  import time
3  N = 3
4  forks = [threading.Lock() for _ in range(N)]
5  def philosopher(i): 1 usage
6      for _ in range(3): # Each philosopher eats 3 times
7          print(f"Philosopher {i} is thinking")
8          time.sleep(1)
9
10         print(f"Philosopher {i} is hungry")
11
12         left = i
13         right = (i + 1) % N
14         first = min(left, right)
15         second = max(left, right)
16         with forks[first]:
17             with forks[second]:
18                 print(f"Philosopher {i} is eating")
19                 time.sleep(2)
20
21         print(f"Philosopher {i} finished eating")
22 threads = []
23 for i in range(N):
24     t = threading.Thread(target=philosopher, args=(i,))
25     threads.append(t)
26     t.start()
27 for t in threads:
28     t.join()
29 print("All philosophers have finished eating.")

```

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Philosopher 0 is thinking
Philosopher 1 is thinking
Philosopher 2 is thinking
Philosopher 0 is hungryPhilosopher 1 is hungry

Philosopher 0 is eating
Philosopher 2 is hungry
Philosopher 0 finished eatingPhilosopher 1 is eating
Philosopher 0 is thinking

Philosopher 0 is hungry
Philosopher 1 finished eatingPhilosopher 2 is eating

Philosopher 1 is thinking
Philosopher 1 is hungry
Philosopher 2 finished eatingPhilosopher 1 is eating

Philosopher 2 is thinking
Philosopher 2 is hungry
Philosopher 1 finished eatingPhilosopher 0 is eating

Philosopher 1 is thinking

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```
Philosopher 1 is thinking
Philosopher 1 is hungry
Philosopher 1 is eating
Philosopher 0 finished eating
Philosopher 0 is thinking
Philosopher 0 is hungry
Philosopher 2 is eating
Philosopher 1 finished eating
Philosopher 2 finished eating
Philosopher 2 is thinking
Philosopher 0 is eating
Philosopher 2 is hungry
Philosopher 0 finished eatingPhilosopher 2 is eating

Philosopher 2 finished eating
All philosophers have finished eating.
```